

Cedarwings SAS Operations — System Protocol

Last updated: 2026-05-20 (post-audit refresh)

This document explains, in plain language, **how the Operations app works end-to-end** — who does what, where the data flows, and how ISO 13485 traceability is preserved. It complements the in-app help (■ Help button, EN/FR) which describes each page field by field.

1. What the system is

A single web application (PWA) that runs the manufacturing operations of Cedarwings SAS:

- Bloom orders → Production → Quality Control → Packaging → Shipping
- Inventory of raw materials and consumables (FEFO)
- Suppliers (Approved Supplier List + certifications)
- Quality cluster: Non-Conformities, CAPAs, Customer Feedback, Internal Audits
- Team: clocking, time reporting, roles, permissions
- Continuous traceability: every action is signed and dated

The app is delivered from GitHub Pages; data and authentication live in Supabase (Postgres + Auth + Storage + Edge Functions).

2. Roles & access

Role	Typical user	What they can do
Manager	Director, QM	Everything: create/edit/delete, approve requisitions, close NCs/CAPAs, configure roles & routing.
Quality Manager	QM	Owns NCs, CAPAs, audits, feedback investigations, supplier evaluations.
Production Manager	Production lead	Owns production lots, materials, machines, maintenance.
Quality Controller	QC operator	Runs 16-point inspections, raises NCs on rejection.

Operator	Production team	Clocks in/out, advances steps, sees their own work.
Warehouse	Stockist	Receives lots, raises requisitions, manages inventory items.
Customer Service	CS rep	Logs feedback, follows up, marks resolved.

Roles are stored on `auth.users.user_metadata.role`. The **Roles** page maps each role to the pages and actions allowed; access is enforced on the client (UI hides what's not allowed) **and** at the database via Row-Level Security policies + a `cw_is_manager()` helper.

3. Daily operational flow

Bloom case	■	Production lot	■	16-point QC	■	Packaging	■	Shipping
■		■		■				
■		■		■■■ reject?	■■■ NC report	■	CAPA(s)	
■		■						
■		■■■ raw material deducted from inventory (FEF0)						
■								
	■■■ Customer Feedback (post-market surveillance)							
	■■■ investigation (Lot/UDI, root cause, result)							
	■■■ CAPA required?	■	CAPA register					
	■■■ Vigilance reportable?	■	regulator reference recorded					

Every step records **who did it, when, on which device, against which lot/order** and is visible in the Traceability page.

4. The quality cluster — how the pieces fit

Since the 2026-05-20 audit, the quality cluster is fully separated into four distinct registers, each with its own page and lifecycle. They link to each other but do not overwrite each other.

4.1 Non-Conformity (NC)

- One report per defect / process deviation.
- Captures: detection, description, immediate correction (quarantine, scrap...), root cause, severity, status.
- Closes only when *every* linked CAPA is Verified or Closed.
- New tile: ■ **Linked CAPAs** (create inline or pick from register).

4.2 CAPA (Corrective & Preventive Action)

- Lives in its own register (`capa.html`).
- Each CAPA has: source (NC / Feedback / Audit / Other), problem statement, root cause, action plan, owner, due date, status, effectiveness check, verified by/at, closed at.

- Lifecycle: **Open** → **In Progress** → **Verified** → **Closed** (or *Cancelled*).
- CSV export, deep-linkable, status KPIs, source + owner filters.

4.3 Customer Feedback (Réclamations)

- Every complaint, return, warranty claim, suggestion, positive feedback.
- New **Investigation & Compliance** section: Lot/UDI, investigation start, investigation result, root cause, CAPA flag + linked CAPA, vigilance reportable + regulator reference.
- Category-based auto-routing (e.g. *Product Quality* → QM role).
- Read receipts (■/■), follow-up reminders (red/amber/green), Dashboard satisfaction score on closed records (30-day rolling).

4.4 Internal Audit

- Plan and record internal quality audits.
- Each audit can spawn one or more CAPAs (via source = audit).

5. Production & Quality Control

1. **Bloom Import** pulls cases from bloomaligner.fr (CSV / Excel / API).
2. **Production** logs the lot: order, machine, material lot (auto-picked FEFO), operator, quantities upper/lower, status pending → in progress → completed.
3. **Clocking** advances the case step by step (Printing → Thermoforming → Trimming → Packaging ...). Each transition is timestamped and attributed.
4. **Quality Control** runs the 16-point ISO inspection. The controller signs (name + login), submits, and either:
 5. ■ Approves → lot released for shipping
 6. ■ Rejects → step reset, NC banner appears with a pre-filled link
 7. ■ Holds for re-verification
8. New QC fields (2026-05 audit): **Linked NC**, **Procedure reference**, **Measures taken** — so the QC certificate documents not just *what was checked* but *which SOP was followed* and *what was done with a rejected lot*.

6. Suppliers & Inventory

6.1 Approved Supplier List (ISO 13485 §7.4)

- No purchase outside the ASL.
- Status: Approved / Conditional / Suspended / Disqualified.
- Annual re-evaluation; LFA shows red flags for evaluations >12 months.
- **Certifications & contracts** with expiry — auto badges:
 - ■ VALID (>30 days)

- ■ SOON (≤ 30 days)
- ■ EXPIRED (past)
- PDF / image attachments stored in a **private** bucket (`supplier-docs`); the app opens them via short-lived **signed URLs** (5 min). Only the original uploader or a manager can delete a file.

6.2 Inventory

- Items catalogue + Lots (FEFO consumption).
- Each item carries a **Storage conditions** string (now a column in the catalogue table — long text truncated, hover for full).
- Stock alerts auto-feed the Dashboard + can auto-create Requisitions.

6.3 Requisitions

- Workflow: pending → approved (manager) → sent → received.
- On receipt a new inventory lot is auto-created with the supplier lot number, linking back to the requisition for traceability.

7. Notifications & follow-up

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- **Customer Feedback**: assignee gets a ■ toast + persistent banner across every page until they open the record (■→■).
 - **Push notifications** (Web Push + VAPID): assignment, chat mention, follow-up reminders. Delivery is tracked in `push_log` (Sent / Delivered / Opened).
 - **Follow-up dates** are surfaced on the Dashboard with a red/amber countdown.
 - **Reminders cron** runs every 15 minutes (`pg_cron`) to wake the push edge function for overdue follow-ups.

8. Per-record traceability (what stays auditable)

Each record across every page carries the same telemetry — visible on the record's detail view:

- `created_by`, `created_at`
- `last_edited_by`, `last_edited_at`
- `last_viewed_by`, `last_viewed_at`
- `notification_seen_at`, `notification_seen_by` (where applicable)
- Status changes are logged in the chat/discussion timeline of the record

Edits to a customer feedback record are hard-restricted by a database trigger (`cf_enforce_edit_owner`) to: the creator, the current assignee/role member, or a manager. Other users get a permission error straight from Postgres — the UI cannot bypass it.

9. Data storage & RLS

- All tables use **Row-Level Security**; the project pattern is "authenticated = trusted" plus targeted owner/role restrictions on sensitive tables (customer_feedback, capa, etc.).
- Private storage buckets: **chat-media** (case discussion attachments) and **supplier-docs** (cert / contract PDFs). Both require auth + serve signed URLs.
- Helper SQL functions live under **public.cw_*** — they are **SECURITY DEFINER** and use a hard search_path to prevent injection.

10. Where to look when something goes wrong

Problem	First place to check
Missing notification	Settings → Push status · push_log table for delivery state
Wrong assignee	Settings → Category / Role routing · the feedback record's assigned_to field
Slow page	Dev Tools network panel · Supabase Logs (mcp__supabase__get_logs)
Can't edit a record	Record's assignee + creator + your role — only those three can edit feedback; managers always can
Login loop on iOS PWA	Nuclear reset: delete app + Settings → Safari → Clear History · localStorage now mirrors the session for cold-reopen
Storage attachment 404	Verify the signed URL has not expired (5 min); regenerate by re-opening the document

11. Help inside the app

Every page has a ■ floating button in the topbar:

1. Click it.
2. The panel opens with **Role · Purpose · Process · Buttons** for that page.
3. Top-right toggle switches **FR ■ EN** — both languages are kept in sync inside **lang.js**.

After the 2026-05-20 audit the help content was updated to cover: - The new Investigation & Compliance section in Customer Feedback - The new NC link / Procedure / Measures fields in Quality Control - The Linked CAPAs tile in Non-Conformity - The Certifications & contracts section + PDF attachments in Suppliers - The Storage column in Inventory - The brand-new CAPA register page

12. Change history (relevant to this protocol)

Date	What changed	Migration / commit
2026-05-20	Audit module extensions: customer_feedback fields, QC fields, supplier_documents, capa table	20260520140000_audit_module_extensions.
2026-05-20	Private bucket supplier-docs + storage policies	20260520150000_supplier_docs_storage.
2026-05-20	UI rollout (5 pages updated, capa.html added, 11 sidebars get CAPA link)	branch claude/add-case-category-layout-AMA0d
2026-05-20	Help content refreshed in lang.js (FR + EN)	this commit

For field-level descriptions, open the page in the app and click the ■ Help button — the in-app help is always the single source of truth, in both languages.